INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

End Semester Examination, December - 2024

Degree: **B.C.A**, Semester: **Third**

Stream: **BCA**

Paper Code: **BCACC302**

Paper Name: **Object Oriented Programming With C++**

Full Marks:100 Duration:3 Hours

Part - A

Attempt any 10 questions out of 15 questions

Each question carries 2 Marks (2 x 10)

1. Describe how data are shared by functions in a procedure-oriented program. 2

2. How does object-oriented approach differ from object-based approach? 2

3. Find errors, if any, in the following C++ statements: 2
   a) cout << "x = "x;
   b) m = 5; / / n = 10; / / s = m + n;
   c) cin >> x; >> y;
   d) cout << \ n "Name:" << name;
   e) cout << "Enter value:"; cin >> x;
   f) / * Addition */ z = x + y;

4. Write a program to read the values of a, b and c and display the value of x, where x = a / b – c 2
   Test your program for the following values;
   a) a = 250, b = 85, c = 25
   b) a = 300, b = 70, c = 70

5. Why does C++ have type modifiers? 2

6. In C++, a variable can be declared anywhere in the scope. What is the significance of this feature? 2

7. What is the application of the scope resolution operator :: in C++? 2

8. The effect of a default argument can be alternatively achieved by overloading. Discuss with an example. 2

9. Illustrate a C++ program to calculate the area and perimeter of a rectangle as well as volume of a cube using the concept of class and object. The input value will be passes when the object calls the member function. Analyse. 2

10. What is a constructor? Is it mandatory to use constructors in a class? 2

11. Interpret the meaning of polymorphism in C++ language. 2

12. Distinguish between early binding and late binding. 2

13. Justify words such as cin and cout not considered as key words. 2

14. Discuss the implications of size parameter in the following statement: 2
cout.write(line, size);

15. Distinguish between overloaded functions and function templates. 2

Part - B
Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 x 6)

16. Describe inheritance as applied to OOP. 5

17. Can we assign a **void** pointer to an **int** type pointer? If not, why? How can we achieve this? 5

18. Find errors, if any, in the following function prototypes. 5
    a) float averages (x, y);
    b) int mul (int a, b);
    c) int display (...);
    d) void Vect (int? & V, int & size);
    e) void print (float data [], size = 20);

19. Develop an application where requirement is to search an item from two different types of data set of 10 elements each. One data set is of type integer and another is of type float. These data needs protection for unauthorized users. Function name of the search function need to be same. 5
Test case:
Input:
Enter integer data set: 2, 5, 4, 3, 7, -9, 10, 12, 45, 17
Enter float data set: 12.4, 2.1, 5.4, 4.6, 1.4, 4.2, 5.4, 6.9, 9.8, 9.7
Output:
Enter integer data to search: -9
Data found at position: 6
Enter float data to search: 9.01
Data not found

20. Identify the error in the following program. 5

```
#include <iostream.h>
Class Room
{
        int width;
        int height;
        static int copyConsCount;
public:
        void room ()
        {
                Width = 12;
                Height = 8;
        }
        Room (Room & r)
        {
                Width = r. width;
                Height = r. height;
                copyConsCount++;
{
        Void dispCopyConsCount ()
        {
           cout << copyConsCount;
        }
};
int room: copyConsCount = 0;
void main ()
{
         Room objRoom1;
         Room objRoom2(objRoom1);
         Room objRoom3 = objRoom1;
         Room objRoom4;
         objRoom4 = objRoom3;
         objRoom4.dispCopyConsCount ();
}
```

21. Identify the error in the following program. 5

```
#include <iostream.h>
Enum weekdays
{
mSunday,
mMonday,
mTuesday,
mWednesday,
mThursday,
mFriday,
```

```
mSaturday,
};
bool op= = (WeekDays & w1, WeekDays & w2)
}
if (w1 = = mSunday && w2 ==mSunday)
        return 1;
else if (w1= = msunday && w2 == mSunday)
        return 1;
else if (w1= = msunday && w2 == mSunday)
        return 1;
else if (w1= = msunday && w2 == mSunday)
        return 1;
else if (w1= = msunday && w2 == mSunday)
        return 1;
else if (w1= = msunday && w2 == mSunday)
        return 1;
else if (w1= = msunday && w2 == mSunday)
        return 1;
else
        return 0;

}
void main ()
{
        Weekdays w1 = mSunday, w2 = mSunday;
        If (w1==w2)
            cout << "Same day";
        else
            cout << "different day";
}
```

22 Write down a C++ program to store and display the details (such as employee id, name and salary) information of required employee of an organization using the concept of method, class and objects. At the same time security of the data set is also essential. The user must define the total number of employees which will be store and display the data as a tabular format. 5

Test case:

Input:
The total number Employee: 2
Details of Employee 1
Employee ID: 201
Name: Rohit Kumar Das
Salary; 45000
Details of Employee 2
Employee ID: 202
Name: Amitava Dutta

Salary: 55000
Output:
201 Rohit Kumar Das 45000
202 Amitava Dutta 55000

23. An institute wants to calculate the total salary of HOD and Assistant HOD. HOD get 60% DA and 30% HRA of basic salary on the other hand Assistant HOD get 50% DA and 20% HRA of basic salary. For both type of employee 12% P. Tax will be deducted from their total salary. To solve this problem, we must implement the concept of friend function. 5

Test case:
Input:
Basic of HOD: Rs.25000/-
Basic of Assistant HOD: 20000/-
Output:
Total salary of HOD: Rs.41800/-
Total salary of Assistant HOD: Rs.28160/-
Total salary of both employee: Rs.69960/-

24. A pointer of base class type points to an object of the derived class. Justify your answer with suitable example. 5

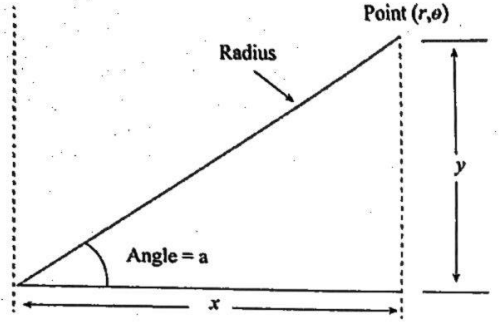
Part - C

Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 x 5)

25. Distinguish between the following terms: 10
    a) Objects and classes
    b) Data abstraction and data encapsulation
    c) Inheritance and polymorphism
    d) Dynamic binding and message passing

26. State whether the following statements are TRUE or FALSE. 10
    a) Data items in a class must always be private.
    b) A function designed as private is accessible only to member functions of that class.
    c) A function designed as public can be accessed like any other ordinary functions.
    d) Member functions defined inside a class specifier become inline functions by default.
    e) Classes can bring together all aspects of an entity in one place.
    f) Class members are public by default.
    g) Friend functions have access to only public members of a class.
    h) An entire class can be made a friend of another class.
    i) Functions cannot return class objects.
    j) Data members can be initialized inside class specifier.

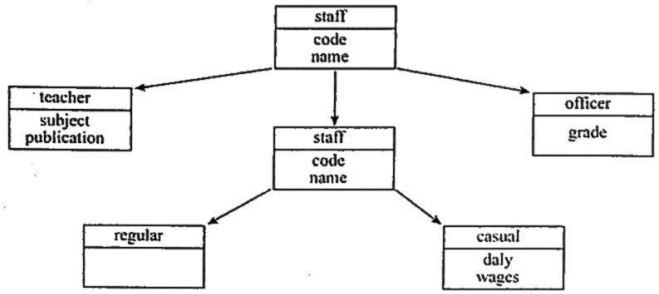
27. Design a class **Polar** which describes a point in the place using polar coordinates **radius and angle.** A point in polar coordinates is shown in figure below. 10



Use the overloaded operator to add two objects of polar.
Note that we cannot add polar values of two points directly. This requires first the conversion of points into rectangular coordinates, then adding the corresponding rectangular co-ordinates and finally converting the result back into polar co-ordinates. You need to use the following trigonometric formulae:

```
x = r * cos (a);
y = r * sin (a);
a = atan (y / x);        // are tangent
r = sqrt (x * x = y * y);
```

28. An educational institution wishes to maintain a database of its employees. The database is divided into a number of classes whose hierarchical relationships are shown in figure below. The figure also shows the minimum information required for each class. Specify all the classes and define functions to create the database and retrieve individual information as and when required. 10



29. An electronics shop, ABC provide a Diwali discount for different product on the basis of MRP range, they provide two types of discounts (flat and special) for some products, the list is given below. 10

Sl No	Product name	MRP Range (Rs.)	Flat Discount	Special Discount
1	TV	>= 20000	15%	10%
		<20000 & >= 15000	10%	5%
		<15000	5%	Nil
2	AC	>=40000	20%	15%
		<40000 & >= 35000	15%	10%
		<35000 & >= 30000	10%	5%
		<30000	5%	Nil
3	Computer	>=35000	10%	7%
		<35000 & >= 25000	7%	5%
		<25000	5%	Nil
4	Others electronics goods	NA	5%	Nil

Now write down a C++ program to calculate the selling price of any product as per user choice based on the concept object and class. For security reasons, the data must be encapsulated

Test case:

Entry your choice:
1. TV
2. AC
3. COMPUTER
4. OTHER

Input:
2
Price of AC:
30000
Output: Selling price of AC is: 24000

30. An organization holds 4 different types of employee (manager, developer and accountant and group-D staff). The salary brake up are given bellow: 10

Emp. Type	Basic	DA	HRA	TA	Medical
Manager	40000.50	16000.2(40% of basic)	6000 (15% of basic)	1000	--------
Developer	30000	12000 (40% of basic)	4500 (15% of basic)	500	--------
Accountant	20000	10000	3000	--------	--------
Group-D	10000.5	--------	--------	--------	500

Write down a C++ program to calculate the total salary of each employee along with brake up salary. To solve this problem, you must implement the concept of function overloading.

Test case:

Basic of Manager is: 40000.50
DA of Manager is: 16000.20
HRA of Manager is: 6000
TA of Manager is: 1000
Total salary of Manager is: 63000

Basic of Developer is: 30000
DA of Developer is: 12000
HRA of Developer is: 4500
TA of Developer is: 5000
Total salary of Developer is: 47000

Basic of Accountant is: 20000
DA of Accountant is: 10000
HRA of Accountant is: 3000
TA of Accountant is: 5000
Total salary of Accountant is: 33000

Basic of Group-D: 10000.50
Medical of Group-D: 500
Total salary of Group-D is: 10500

C++ problem based on operator overloading

31. An organization recruits either engineer or MBA pass outs. For engineer, the organization store, information such as engineering stream, project code, and total salary plus an incentive of Rs: 9000. For MBA graduates, the organization keeps information such as CAT score, MBA institute, and specialization incentive is Rs: 10000. Now the organization plans to recruit candidates who have done engineering as well as MBA. For those candidates, organization needs to keep all above details as well as a ratio of job responsibility % such as engineering: 65%, management 35% incentive will be based on % of job responsibility. Develop all three applications using object-oriented programming concepts also display details of a BTech & MBA candidate. **10**

32. A mathematical research institute needs to implement stack. But requirement is such that for different time (execution) they need stack of different types. For example, when integer type selected the stack and item should be of type int. same is applicable for float and character type. Implement the above requirement by object-oriented programming. **10**

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